

SYLLABUS: ECE 315 – SPRING 2026

Course Name: Signals and Systems

Lecture Time: Mon, Wed, Fri, 10:20 am – 11:10 am

Lecture Location: Min Kao Building 404

Instructor: Dongsheng Ding (dongshed@utk.edu)

Office: Min Kao Building 315

Office Hours: Mon, Wed, 11:10 am – 12:30 pm



MIN H. KAO DEPARTMENT OF
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Course Description: This course introduces continuous- and discrete-time functions, function transformations, signal energy and power, solution of linear differential equations, system properties, convolution, continuous and discrete-time Fourier series, continuous and discrete-time Fourier transforms, Bode diagrams, and correlation.

Prerequisites: Familiarity with undergraduate-level calculus, linear algebra, and probability is recommended.

Recommended Materials

- **(SS)** Signals and Systems, 2nd Edition, A. Oppenheim, A. Willsky, S. Nawab, Prentice Hall, 1997.
- **(LSS)** Linear Systems and Signals, 3rd Edition, B.P. Lathi, R. Green, Oxford University Press, 2017.

SS is the main reference for the class. Both will be used to assign readings and homework problems.

Course Website: CANVAS in your myutk account.

Teaching Assistants:

- Xinyu Feng (xfeng15@vols.utk.edu)

Office and office hours: Min Kao Building 213, Tue, Thu, 1:30 pm – 2:30 pm

- Atik Rahman (arahma19@vols.utk.edu)

Office and office hours: Min Kao Building 540, Tue, Thu, 11:05 am – 12:05 pm

Course Outline

- Basics of discrete- and continuous-time signals (1 week)

Notation: discrete vs continuous, signal size, signal operation, periodicity, symmetry, examples: exponential, sinusoidal, impulse, step functions

- Basics of discrete- and continuous-time systems (1 week)

Interconnection, system properties: memoryless, invertibility, causality, stability, time-invariance, linearity

- Discrete-time linear time-invariant systems (2 weeks)

Convolution and its properties, impulse response, step response, system stability, block diagram

- Continuous-time linear time-invariant systems (1 week)
Convolution and its properties, impulse response, step response, system stability, block diagram
- Fourier series of continuous-time periodic signals (2 weeks)
Representation, properties of Fourier series, Parseval's theorem
- Fourier series of discrete-time periodic signals (1 week)
Representation, properties of Fourier series, Parseval's theorem
- Continuous-time Fourier transform (2 weeks)
Representation, existence and convergence, properties of Fourier series, Parseval's theorem
- Discrete-time Fourier transform (1 week)
Representation, existence and convergence, properties of Fourier series, Parseval's theorem
- Time and frequency characterization of signals and systems (2 weeks)
Magnitude-phase representation, first-order and second-order continuous- and discrete-time systems.

Grading: The final grade consists of 5% Participation, 20% Homework, 20% Midterm Exam I, 20% Midterm exam II, and 35% Final Exam.

- Participation includes taking and sharing lecture notes, asking questions during class, and providing constructive feedback.
- Homework consists of approximately 10 weekly assignments, all equally weighted. Discussion with others is allowed; however, all solutions must be hand-written independently (i.e., independent human intelligence), scanned into a single PDF, and submitted on CANVAS. Please ensure that your PDF is clearly readable on a computer. Homework is usually due one week after it is assigned, unless otherwise specified. **Late submissions will not be accepted.** Your lowest homework score will not be counted.
- Homework may include problems intended to practice coding. While your solutions will be reviewed, they will not be graded. Python or MATLAB is acceptable.
- There will be two in-class midterm exams and one final exam that is more comprehensive. We estimate that the first midterm will be in **February 20** and the second midterm will be in **April 10**; these will be confirmed one week before the exam date. The final exam will be on **May 12**; see <https://registrar.utk.edu/calendars-and-catalog/>.

Letter grades will be assigned using the following scale:

A, A-	[90, 100], [87, 90)
B+, B, B-	[83, 87), [80, 83), [77, 80)
C+, C, C-	[73, 77), [70, 73), [67, 70)
D+, D, D-	[63, 67), [60, 63), [57, 60)
F	[0, 57)

This scale may be adjusted depending the overall performance of the class. Any adjustment will only lower the cutoffs to shift the grade distribution in students' favor.

Use of Generative AI: Any use of generative AI tools (e.g., ChatGPT, Gemini, Copilot) must be explicitly acknowledged.

- Directly copying and pasting the content generated by generative AI tools is prohibited. We will be actively looking into the abuse of generative AI tools, which might be reported as academic misconduct. If you are not sure whether the output of a generative AI tool is permitted aid in the context of a particular assignment, please talk to us.
- You may use generative AI tools to assist with your homework. For example, you may ask generative AI tools to check your solutions or ideas and to generate initial Python or MATLAB code. If you do so, your homework submission must include which generative AI tools, your prompts, the generated content, and the date and time of access, along with a detailed explanation of how these results were used to arrive at your solutions (line by line). You are responsible for the correctness of all reasoning steps and final answers.
- At any point during this semester, when the instructor and or TA's suspect that your use of generative AI tools has negatively affected your independent understanding of the course material, you will be invited to present your solutions in person during office hours. If your presentation cannot demonstrate that your solutions were completed independently with human intelligence, your grade might be reduced or dismissed accordingly.

Electronic Devices: During exams, you may use simple scientific calculators that do not have wireless communication. Laptops, cell phones, tablets, and any other electronic devices capable of communication must be turned off during the exam.

Reggrading: After homework and exams are graded and returned, you will have one week to dispute the grading if you believe it is incorrect. To do so, you must send an email to TA explaining the reason for your request. Please note that your work will be re-assessed, and your grade may be adjusted either upward or downward.

Study Center: The Tickle College of Engineering Study Center, located in Ferris 334, assists students with their more challenging and/or upper-level engineering courses. Students can book appointments online or drop-in during tutoring hours. Information for which courses are available and when can be found at <https://tickle.utk.edu/student-experience/student-success/study-center/>. The TCE Study Center will open on Monday, January 26th, and close on the last day of classes.

ADA and Accessibility: If you need any accommodation, please let me know as soon as possible. Please find the university services at <https://sds.utk.edu>

Honor Statement: All students are subject to the Student Conduct & Community Standards: <https://studentlife.utk.edu/studentconduct/academic-dishonesty/>